

2023



i GRID LAB Builds AI-Controlled Platform for Excess Power with the Goal of Circulation Green Power Regionally on AWS

Businesses building energy platforms using AI, IoT, digital, and cloud technologies

[Overview](#) | [Opportunity](#) | [Solution](#) | [Architecture](#) | [Outcome](#) | [AWS Services Used](#)

6 months

Time needed to start the demonstrati



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1 month

Successfully achieved demonstration r



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New functions can be made stable and operational in a short period of time.

Overview

[i GRID LAB](#) conducts research and develops solutions for decarbonization through green tech. In June 2022, the company released R.E.A.L. New Energy Platform, which analyzes a wide variety of data using AI and performs detailed control of surplus solar power. By adopting Amazon Web Services (AWS) as the foundation of its platform, the company has achieved a proof of concept in a short period of time and high-speed processing of huge amounts of data, contributing to the acceleration of green transformation (GX).



Opportunity | Expanding Renewable Energy Use with AI

i GRID Solutions and its group companies are developing its business centered on Green Transformation (GX) Solutions. It promotes GX in order to accelerate its adoption by making social and economic activities centered on renewable energy and decarbonization a reality



In June 2022, i GRID LAB released the R.E.A.L New Energy Platform and started full-scale deployment to a group company, VPP Japan. The R.E.A.L. New Energy Platform is an energy management platform that achieves highly accurate surplus power prediction through artificial intelligence (AI) and detailed automatic control of Distributed Energy Resources (DER). It collects information on things such as solar power generation volume and power, demand, storage batteries, and EV charging/discharging status, and the data is stored in the cloud. AI performs highly accurate predictive analysis combined with external information such as electricity market prices. AI is also used to determine the optimal timing for EV charging and discharging along with the calculation of excess solar energy and the creation automatic device control scenarios.

VPP Japan owns R.E.A.L. Solar Power, one of the largest distributed solar power plant in Japan, and the first goal is to use this platform to circulate the excess power that is generated by the plant regionally.

Tetsu Iwasaki, director and chief technology officer (CTO) of i GRID LAB, says, "We thought that using AI would be an important key to expand the use of renewable energy." The number of self-consumption solar power systems in places like commercial facilities is increasing, and it is difficult to control excess power. Until now, it has been common to only be able to get the amount of power that a facility could fully consume. "The basic concept is to properly control the surplus power with AI and provide it to the other users. This makes it possible to achieve a circulation model for surplus power and increase the utilization ratio of renewable energy by maximizing the introduction of solar power generation."

Solution | Building a Proof of Concept in only 6 Months after Project Launch, with First Results Published 1 Month Later

Iwasaki, who joined i GRID LAB in October 2020, began a full-scale project to build a next-generation energy platform in December. He started development on AWS in January 2021. Meanwhile, he asked for help from supermarket chains that had already installed solar panels at multiple locations under the i GRID Group. A proof of concept was conducted at two stores in Saitama prefecture."Building such a platform is the first initiative in the industry, and by actually creating and running it, rather than studying it on a desk, we can concretely ascertain the effects and problems. It was important for us to be agile and start moving quickly," says Iwasaki.

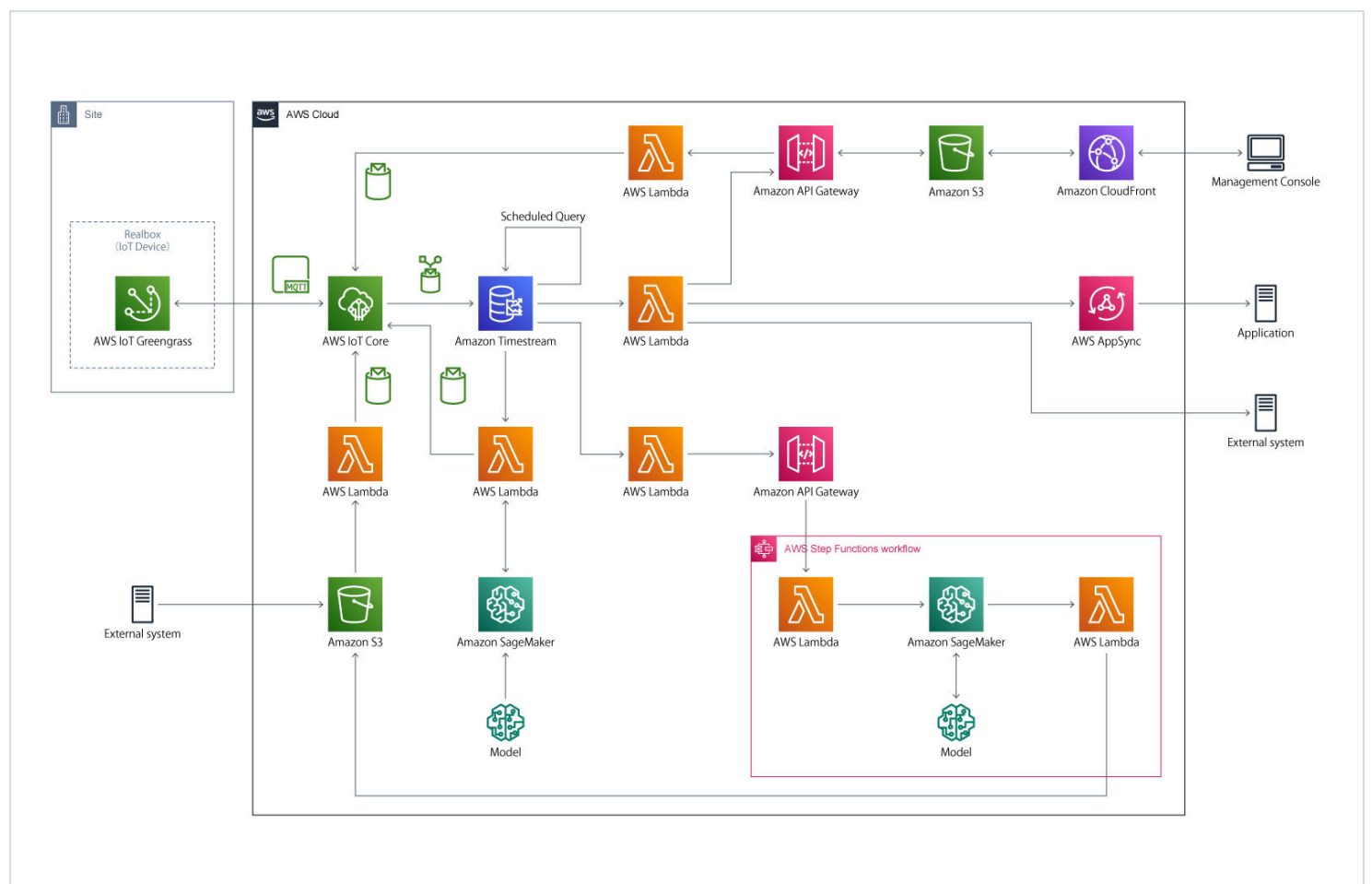


A platform for the proof of concept was built in just 6 months from the start of the project. The company adopted [AWS IoT Greengrass](#) for the data collection from edge devices and [Amazon Timestream](#) for high-speed processing of large amounts of time series data and began proof of concept in June 2021. In July, they clarified the specific effects and problems and announced the details on their website. Additionally, they migrated AI functions, which had been operated on external servers until then, to [Amazon SageMaker](#).

They continued the proof of concept after that and, as a second step, commenced the decarbonization of last mile logistics, which incorporates EV for online supermarkets, in January 2022. At the same time, the i GRID Group began providing this adaptation to around 500 solar facilities. Furthermore, in December 2021, they collaborated with a delivery system for sending power to general households. They collected and analyzed electricity usage and other data from approximately 100,000 households.

And, as mentioned previously, the platform was commercialized in June 2022. The brand names of related services and products have been unified into R.E.A.L., and mutual cooperation between various services provided by the i GRID Group has also been strengthened.

Architecture



Outcome | Unifying Cloud Configurations with AWS

Iwasaki says they chose AWS for the R.E.A.L. New Energy Platform foundation because of three main reasons. First, AWS provides serverless services, which allow development to proceed quickly. "It has been said that the world has already entered into the VUCA (Volatility/Uncertainty/Complexity/Ambiguity) era, and even v... energy industry continues to change at a rapid pace. Minimizing the duration of coding and document... in order to continue to respond quickly to this changing era. This can be achieved easily by leveraging the serverless

capabilities of AWS. Even in a proof of concept, it is now possible to perform a verification on a monthly cycle through making detailed releases on a weekly basis.”

Second, AWS can process large amounts of data at high speeds. Building on AWS, we collect time series data from various sensors on edge devices every minute, and weather information data and power market price data moment by moment. As the number of managed power generation equipment and receivers increases, the amount of data increases proportionally. “Since the i GRID’s business is expanding at a very high speed, there is a possibility that the scale of operation will be larger than originally planned, but with AWS, which has excellent scalability, we can operate with peace of mind.”

Third, it is easy to secure engineers familiar with building on AWS and to receive technical support from AWS teams. “Since we would like to implement a lot of features for the R.E.A.L. New Energy Platform, it is also important that it is easy to set up a development team that can accommodate this. And also, AWS can respond quickly when problems occur, and new features can be run stably in a short period of time. At the beginning, we also considered a multi-cloud configuration, but finally, we decided that integrating into AWS would have greater benefits,” adds Iwasaki.

Currently, the company is cooperating with the commercial facilities as well as local governments and regional banks in order to expand the use of the R.E.A.L. New Energy Platform. They aim to achieve a GX city on the surface by making the entire city’s energy 100 percent renewable. “Until 2030, we aim to build 10 GX cities with 100 percent renewable energy using this platform,” says Iwasaki. He also has a plan to revitalize regions through decarbonization and connecting it to economic growth.

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About i GRID LAB

A subsidiary of i GRID Solutions, which handles DER (distributed energy resources) platform construction, AI-powered energy management service “Enudge,” and power retail (for corporations/households), i GRID LAB carried out the construction of R.E.A.L. New Energy Platform, a distribution energy platform that utilizes AI, IoT, and cloud technology, analyzes big data, and provides support for the use of AI and DX.



Tetsu Iwasaki

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2

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